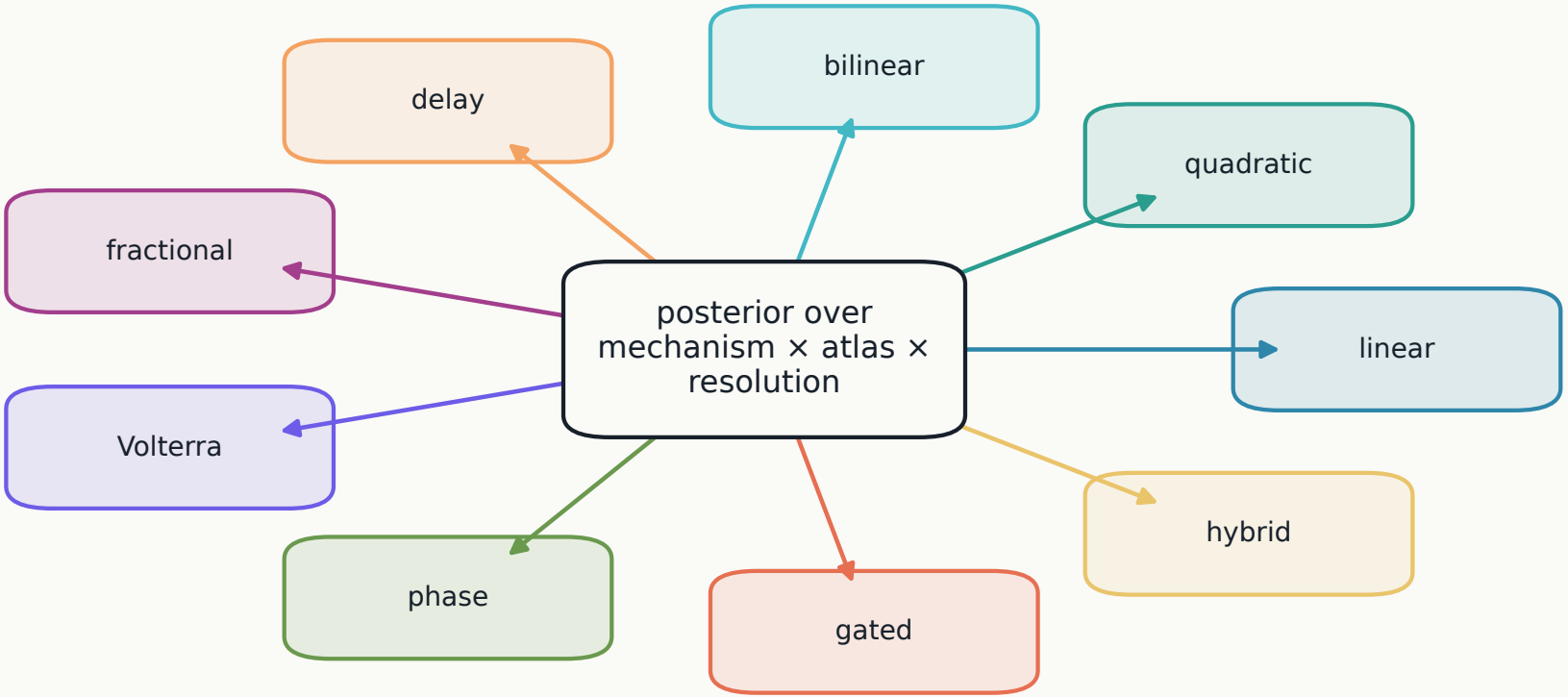Typed operator-program grammar